A

MINI PROJECT REPORT ON

# Online Food Processing System

**SUBMITTED TO SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE**

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF DEGREE OF

**BACHELOR OF TECHNOLOGY (B.TECH)**

***(Artificial Intelligence & Data Science)***

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***Academic Year: 2026 – 2027***

# Dr. D. Y. Patil College of Engineering and Innovation, Varale

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### CERTIFICATE

This is to certify that the Mini Project entitled

"Online Food Processing System"

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is a bonafide work carried out by them under the guidance of **Mrs. Mayuri Fegade**, in partial fulfillment of the requirements of Savitribai Phule Pune University, Pune for the award of the degree of Bachelor of Technology in Artificial Intelligence and Machine Learning .

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# ACKNOWLEDGENT

It is our pleasure to acknowledge with gratitude all those who helped us complete this mini project. We are highly indebted to **Mrs. Mayuri Fegade** for her guidance, supervision, and valuable suggestions throughout the work. We express our gratitude to **Dr. Dipannita Mondal**, Head of the Artificial Intelligence - Data Science and Machine Learning Department, for her encouragement and for providing the required facilities. We also thank our teachers, friends, and everyone who directly or indirectly supported the completion of this mini project.

Name of the Student and Sign

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# Chapter 1: Introduction

The purpose of this mini project is to design an Online Food Processing System that simplifies the process of browsing food items, placing orders, and managing their preparation. Through the proposed system, we will:

- Provide a digital menu with food descriptions, prices, and availability.
- Allow customers to register, sign in, place orders, and select payment and delivery options.
- Help restaurant staff receive orders and update preparation and order status.
- Provide customers with order confirmation, status tracking, and a clear summary of their purchases.

Background: Online food platforms have changed how customers discover restaurants and order meals. A web-based system can bring menu management, order handling, and customer communication into one convenient interface.

System Context: Customers may otherwise need to call restaurants, repeat order details, or wait for manual confirmation. A centralized online workflow reduces these communication gaps.

Technical Primer: The system can use HTML, CSS, and JavaScript for the user interface, a server-side application for business logic, and a database to store users, menus, orders, and payment status.

Practical Impact: A well-organized online food processing workflow can improve order accuracy, provide timely status updates, and help restaurant staff prioritize incoming orders.

# Chapter 2: Problem Statement

Traditional food ordering methods can involve delays and mistakes because orders are recorded manually or communicated across separate channels. Common issues include:

- Customers may have difficulty viewing current menus, prices, and item availability.
- Telephone or handwritten orders can contain incorrect items, quantities, or delivery details.
- Customers may not receive timely confirmation or updates about preparation and delivery.

This project aims to:

1. Develop a user-friendly website for browsing menus and placing food orders.
2. Provide restaurant staff with an interface to manage incoming orders.
3. Maintain order records and display clear order-status updates to customers.

- Operational Challenges:
    - Menu Management: Changes to item prices or availability must be reflected accurately for customers.
    - Order Coordination: Staff need a consistent way to review, accept, prepare, and complete each order.

- Data Management: Customer details, order items, totals, and statuses should be stored consistently and retrieved when needed.
- Usability: The interface should be simple to navigate on desktop and mobile devices and should provide clear feedback at each step.

# Chapter 3: Objectives

## 1. Build the Customer Interface:

- Display food categories, item descriptions, prices, and availability.
- Provide registration, login, cart, address, and order-confirmation features.

## 2. Implement Order Processing:

- Validate orders and calculate item totals, quantities, and applicable charges.
- Allow staff to accept orders and update preparation and fulfillment status.

## 3. Manage Data Reliably:

- Store customer, menu, and order information in a structured database.
- Check inputs and handle unavailable items or incomplete order details.

## 4. Additional Technical Goals:

- Provide order search and filtering by status, date, or order identifier.
- Include role-based access for customers and restaurant administrators.

## 5. Improve Customer Experience:

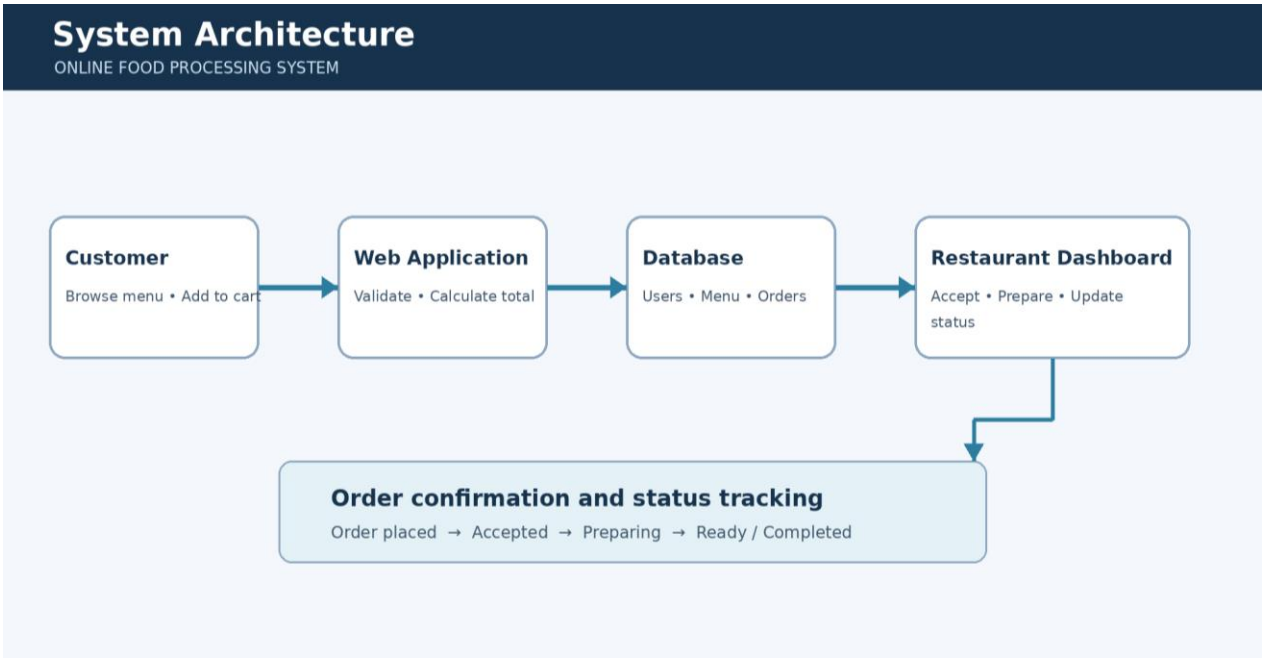
- Provide clear confirmation messages and progress updates after an order is placed.
- Design the pages to be responsive and accessible on common devices.

# Chapter 4: The motivation for this project stems from the growing use of digital services in everyday food ordering:

The motivation for this project stems from three critical realities in today's digital landscape:

1. **Convenience and Speed:**
    - Customers can browse the menu and place an order without making a phone call.
    - **A structured cart lets customers review items and quantities before confirming an order.**
    - **Digital order records can reduce confusion between customer requests and kitchen instructions.**
    - **A shared order-status workflow helps staff coordinate preparation, pickup, and delivery.**
2. **Operational Efficiency:**
    - Restaurant staff can view incoming orders in one place instead of relying only on paper notes or calls.
    - Menu availability and prices can be updated centrally to reduce outdated information.
3. **Better Communication:**
    - Order confirmation and status messages help set expectations for customers.
    - This project applies web development and database concepts to a familiar real-world problem and demonstrates how a digital workflow can support both customers and restaurant staff.

## 4. System Architecture:




### Image Description:

The proposed system begins when a customer browses the menu and adds items to a cart. The application validates the order, records it in the database, and sends it to the restaurant dashboard. Staff update the preparation and fulfillment status, which is then displayed to the customer.

# Chapter 5: Scope and Rationale of the Project

## Scope:

- **Functional Boundaries:**
    - Focuses on menu browsing, customer accounts, cart management, order placement, order records, and order-status updates.
    - Includes a basic administrator interface for maintaining menu items and processing incoming orders.
- **Practical Constraints:**
    - The prototype can run locally using sample menu data and a development database for demonstration.
    - Live delivery partner integration, advanced recommendations, and production payment processing are outside the basic prototype scope.

## Rationale:

1. **Practical Learning:**
    - The project demonstrates how user interfaces, server-side validation, and databases work together in a complete application.
2. **Improved Order Accuracy:**
    - A structured cart and order summary help customers review their selections before submission.
3. **Real-World Relevance:**
    - The workflow reflects common restaurant operations and can be extended for cafes, canteens, and small food businesses.

# Chapter 6: Methodological Details

The project can be developed and tested in the following phases:

## 1. Website and Interface Development

- **Technologies Used: HTML, CSS, JavaScript, and a backend framework such as Python Flask or Node.js.**
- **Features:**
    - Customer registration and login.
    - Food menu, item details, shopping cart, and quantity controls.
    - Checkout form with order summary and payment-method selection

## 2. Database and Order Setup

- Create database tables for users, menu items, orders, and order details.
- **Core Data Fields:**
    - Customer profile and contact information.
    - Menu item name, price, category, and availability.
    - Order identifier, quantities, total amount, timestamp, and current status.

## 3. Order Processing Workflow

- **Customer Order Placement:**
    - Validate customer input and cart contents on the server.
    - Steps:
        1. Customer selects food items and quantities from the menu.The system calculates the total and collects delivery or pickup details.
        2. The order is saved and made visible in the restaurant dashboard.
    - **Demonstration: Display the order confirmation and a unique order reference.**

- **Restaurant Order Management:**
    - Tools: Administrator dashboard and database queries.
    - Steps:
        1. Review new orders and confirm acceptance.
        2. Update order status as accepted, preparing, ready, or completed.
    - **Demonstration: Verify that the updated status appears in the customer view.**
- **Input Validation and Error Handling:**
    - Check required fields, quantity limits, item availability, and valid order references.
    - **Demonstration: Show helpful messages for missing details or unavailable menu items.**

# 4. Testing and Evaluation

- **Testing Steps:**
    1. Place sample orders using different menu items and quantities.
    2. Verify that the stored order details match the submitted cart.
    3. Check that staff status changes are reflected correctly.

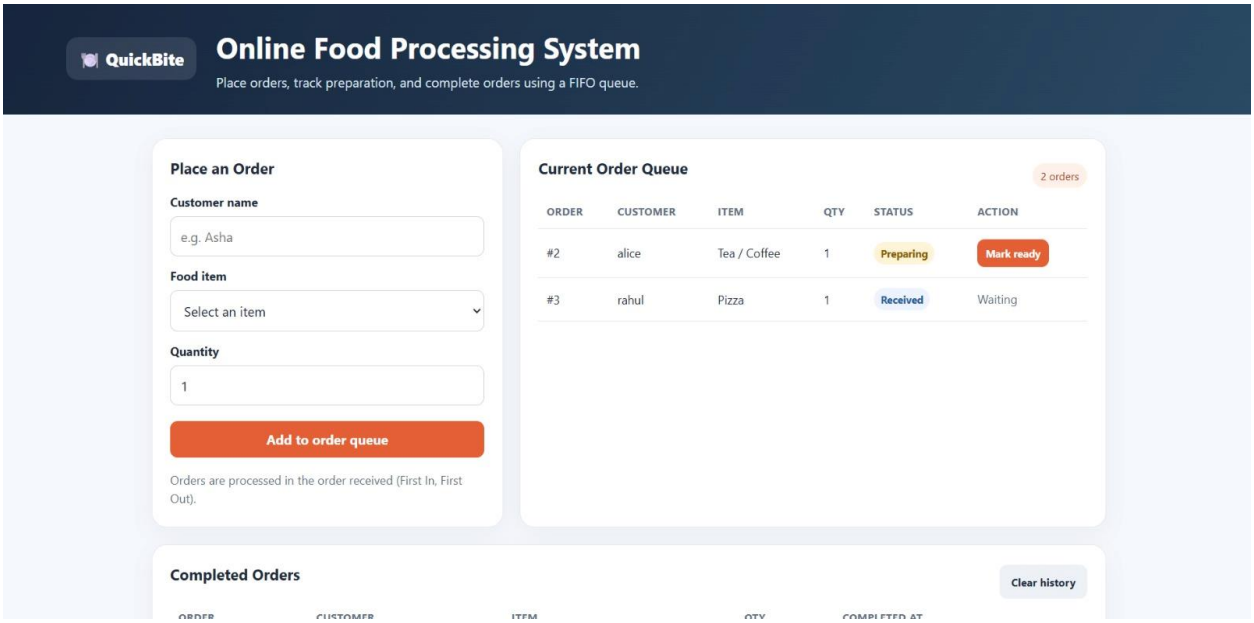
- **Validation:**
    - Test login, menu display, cart calculations, order submission, and order-history retrieval.
    - Repeat test cases for invalid inputs, empty carts, and unavailable items.

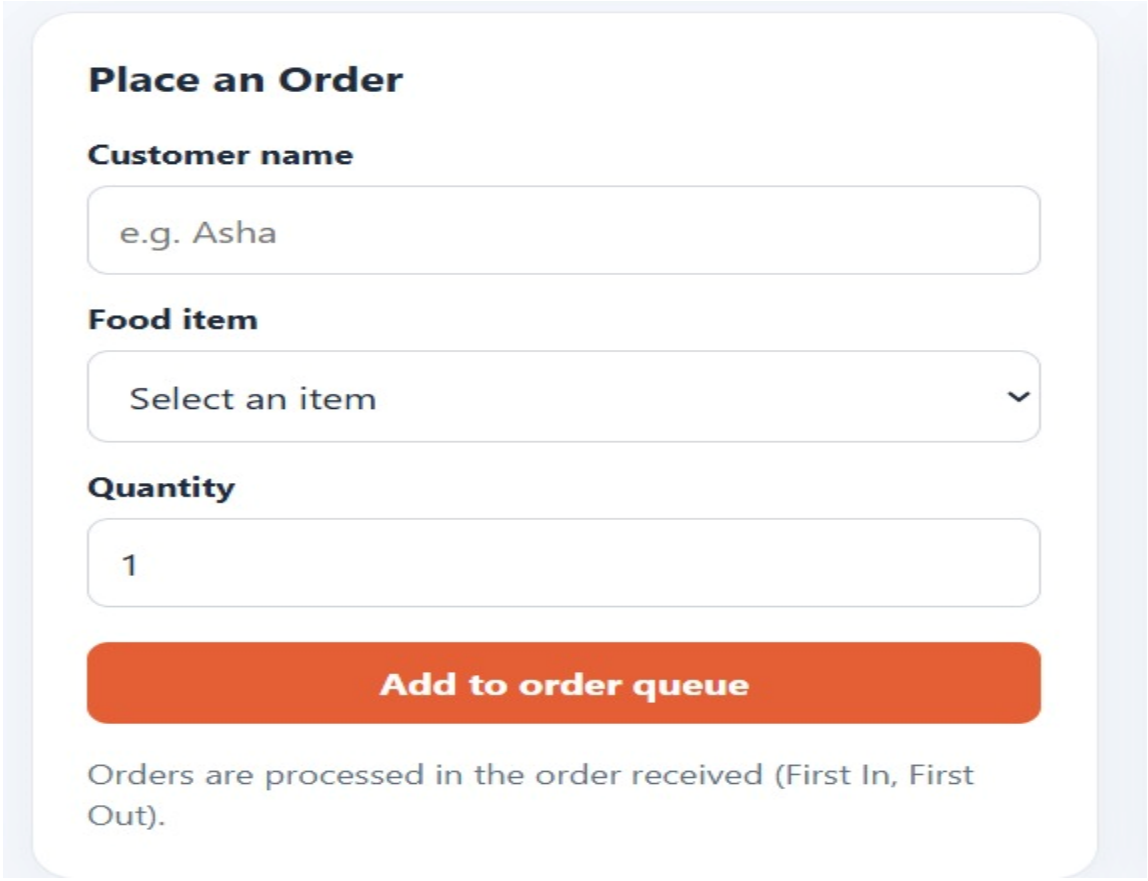
# Chapter 7: Results

| Aspect | Expected / Tested Function | System Output |
|---|---|---|
| **Menu Display** | Food items, prices, and availability shown | Menu details displayed from stored records |
| **Cart Calculation** | Quantities and totals validated | Order summary calculated before submission |
| **Order Storage** | Order and item details saved | Order record available for staff review |
| **Status Updates** | Staff can update order progress | Customer view reflects saved status |
| **Data Exposure**<br>Input Validation | Missing or invalid details rejected | Validation message shown to user |
| **Tampering**<br>**Time**<br>Order History | Previous order details retrievable | Saved orders can be reviewed by reference |
| **XSS Success** | | |

**Key Observations:**

- The prototype workflow supports menu browsing, cart review, and submission of a structured food order.
- Order records and status updates provide a clearer way for customers and restaurant staff to follow an order.


QuickBite
Online Food Processing System
Place orders, track preparation, and complete orders using a FIFO queue.
Place an Order
Customer name
e.g. Asha
Food item
Select an item
Quantity
1
Add to order queue
Orders are processed in the order received (First In, First Out).
Current Order Queue
2 orders
ORDER
CUSTOMER
ITEM
QTY
STATUS
ACTION
#2
alice
Tea / Coffee
1
Preparing
Mark ready
#3
rahul
Pizza
1
Received
Waiting
Completed Orders
Clear history


Place an Order
Customer name
e.g. Asha
Food item
Select an item
Quantity
1
Add to order queue
Orders are processed in the order received (First In, First Out).

## Current Order Queue

2 orders

| ORDER | CUSTOMER | ITEM | QTY | STATUS | ACTION |
|---|---|---|---|---|---|
| #2 | alice | Tea / Coffee | 1 | Preparing | Mark ready |
| #3 | rahul | Pizza | 1 | Received | Waiting |

## Completed Orders

Clear history

| ORDER | CUSTOMER | ITEM | QTY | COMPLETED AT |
|---|---|---|---|---|
| #1 | sam | Veg Sandwich | 4 | 8:50:19 PM |

# Chapter 8: Analysis

Key Findings:

- **Centralized Order Handling:** A single order dashboard helps staff review incoming requests and manage their status.

- **Data Consistency:** The application should validate inputs and use database transactions where appropriate**.**
  - Menu and quantity validation reduce invalid submissions.
  - Server-side total calculation helps prevent reliance on values supplied only by the browser.

- **Usability:** Clear labels, confirmation messages, and responsive pages are important for a smooth ordering experience**.**

- **Functional Testing:**
  - Test cases should verify successful order placement, correct totals, persistence of order records, and status updates.
  - Performance: Response time should be evaluated with realistic menu sizes and multiple simultaneous requests before deployment.

- **Limitations:**
  - The prototype may use sample data and does not necessarily connect to live restaurants or delivery services.
  - Real payment gateways require secure integration, testing, and appropriate compliance measures.

- Operational Considerations: Reliable hosting, database backups, access control, and protection of customer information are needed for a production deployment.

# Chapter 9: Interference

- **Developer Responsibility:**
  Use server-side validation, secure password storage, access controls, and careful error handling.
  Developers should protect customer data and ensure that only authorized staff can modify menu items or order statuses.

- **Business Impact:**
  A centralized workflow can help restaurants organize requests and maintain a searchable record of orders.

- **Testing Inference:**
  Functional tests should confirm that the cart total, saved order, and displayed status remain consistent throughout the workflow.

- Online ordering can improve convenience when supported by accurate menus and dependable order processing.

- A clear and validated workflow reduces avoidable order-entry errors, although staff review and operational procedures remain important.

# Chapter 10: Conclusion

This project presents the design of an Online Food Processing System for customers and restaurant staff.

- It provides a digital menu and a structured way to select food items.
- It organizes order submission, database storage, and preparation-status updates.
- It demonstrates how a web application can make food ordering more convenient and easier to manage.

**Critical Takeaway: A reliable online food processing system combines a simple customer experience with accurate order records and clear restaurant workflows.**

- **Future Enhancements:**
  - Integrate a verified payment gateway and automated payment-status reconciliation.
  - Add delivery tracking, notifications, and reporting for restaurant administrators.
- **Explore inventory integration, customer feedback, and recommendations based on order history.**

# Chapter 11: Suggestions

1. **For Developers:**
    - Validate all important fields on the server and calculate order totals on the backend.
    - Use secure authentication, role-based permissions, and protected database access.
    - Test cart calculations, order creation, and status transitions with positive and negative test cases.
    - Back up the database and log errors without exposing sensitive customer details.
    - Make the interface responsive, accessible, and easy to navigate.
2. **For Restaurants:**
    - Keep menu descriptions, prices, and availability up to date.
    - Train staff to accept orders and update preparation status promptly.
    - Review order records regularly to identify recurring issues and improve service.
    - Plan for peak demand and ensure the system has reliable hosting and connectivity.
3. **For Customers:**
    - Review item quantities, address, and total before confirming an order.
    - Provide accurate contact and delivery information.
    - Use the order reference and status updates to follow the request.
    - Share constructive feedback to help improve the ordering experience.

# Chapter 12: Acknowledgments

I sincerely thank **Dr. Dipannita Mondal**, HOD of AI & ML, for her guidance and encouragement. I am grateful to **Mrs. Mayuri Fegade** for her valuable supervision and suggestions throughout this mini project. I also thank the department and college for providing the facilities required for learning and development.

I appreciate the support of my project teammates for their collaboration in planning, interface development, database design, implementation, and testing. Their contributions helped us understand the complete workflow of an online food processing application.

Special thanks to the authors and maintainers of the documentation and open-source tools used during development, including web technology, backend framework, and database resources.

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